

Acute myeloid leukemia

≥20% myeloid blasts — or any blast % with a defining genetic lesion · classification and treatment are now genetically driven · every new diagnosis is triaged for the three emergencies before a regimen is chosen

Compiled from the Encyclopedia AML review note and Fellowship Anki deck · ELN 2022 · WHO 2022 · ICC · NCCN. Items marked † extend beyond the existing review note.

DIAGNOSIS

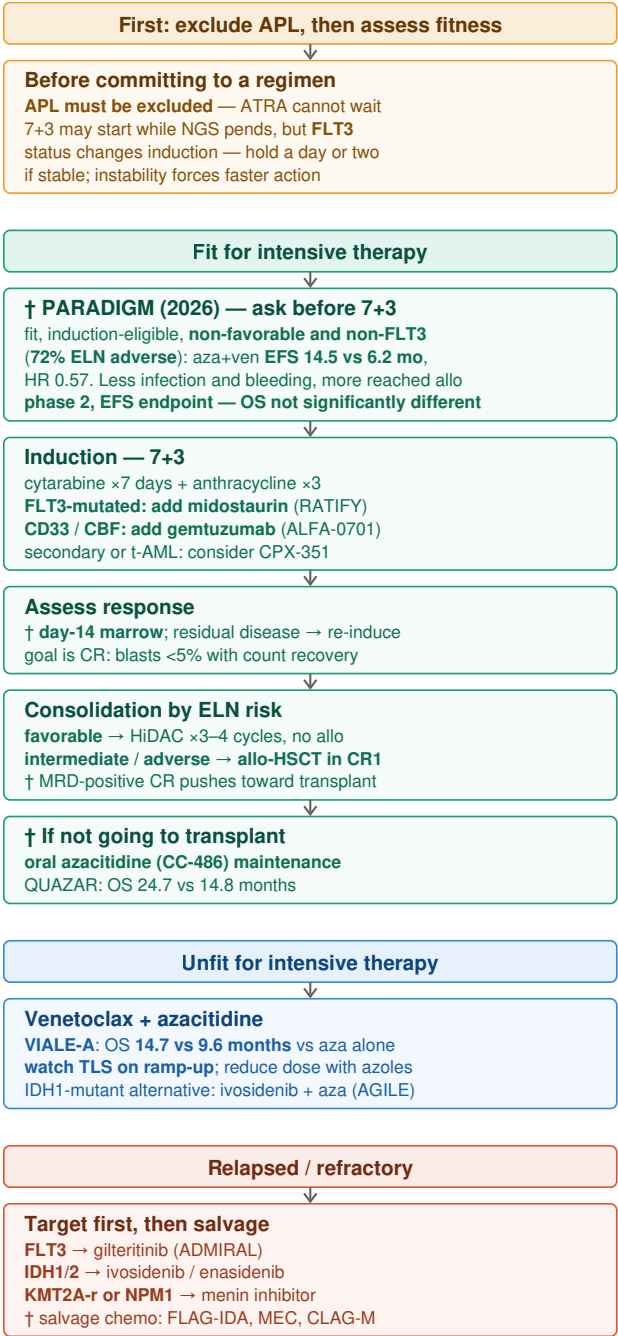
- ≥20% myeloid blasts in blood or marrow — or any blast % with a defining genetic lesion: t(8;21), inv(16)/t(16;16), PML::RARA, and (WHO 2022) NPM1 or CEBPA-bZIP
- Also AML at <20%: **myeloid sarcoma** / extramedullary disease
- “Blasts” on smear is morphologic — **confirm lineage by flow**
- Workup: marrow aspirate + biopsy with **flow, karyotype + FISH, NGS**; CBC/smear (Auer rods); **coags + fibrinogen + D-dimer**; TLS labs; **baseline echo**; HLA typing early; LP only if CNS symptoms or monocytic, after coags corrected
- **Stat FISH/PCR for PML::RARA** on any suspicion of APL
- † Median age ~68. The three triage questions on every new AML: **Is it APL? Is there leukostasis? Is there tumor lysis?**

ELN 2022 GENETIC RISK

Risk	Representative genetics
Favorable	t(8;21) RUNX1::RUNX1T1 · inv(16)/t(16;16) CBFB::MYH11 (core-binding factor) · mutated NPM1 without FLT3-ITD · in-frame bZIP CEBPA
Intermediate	NPM1 + FLT3-ITD · wild-type NPM1 + FLT3-ITD · t(9;11) MLLT3::KMT2A · anything not otherwise classified
Adverse	Complex or monosomal karyotype · -5/del(5q), -7, abn(17p) · TP53 · MDS-related genes (ASXL1, BCOR, EZH2, RUNX1, SF3B1, SRSF2, STAG2, U2AF1, ZRSR2) · KMT2A rearrangements · t(6;9), inv(3)

ELN 2022 changes worth knowing: FLT3-ITD is intermediate regardless of allelic ratio (ratio dropped) · only **in-frame bZIP** CEBPA is favorable · **NPM1** with adverse cytogenetics is reclassified adverse · **KIT** co-mutation worsens core-binding-factor AML.

TREATMENT PATHWAY



REGIMENS & AGENTS

Regimen	Notes
7+3	Cytarabine continuous infusion ×7 days + daunorubicin or idarubicin ×3 days. Daunorubicin 90 mg/m² > 45 in age <60 (E1900); 60 mg/m² when a second agent is added
+ midostaurin	FLT3-mutated (RATIFY). Quizartinib for FLT3-ITD (QuANTUM-First)
+ gemtuzumab	CD33-positive, fractionated dosing; greatest benefit in core-binding factor (ALFA-0701)
CPX-351	Liposomal cytarabine + daunorubicin for t-AML / AML-MRC. Caveat : comparator was 7+3, not ven/aza — niche today is the fitter older adult bridging to allo
HiDAC	3 g/m² q12h ×3–4 cycles; best in CBF. Age >60 → IDAC 1.5 g/m² — 3 g/m² is inappropriate (ocular, cerebellar, infectious toxicity). Renal dose-reduce
Venetoclax + azacitidine	Unfit patients (VIALE-A). Ramp-up with TLS prophylaxis; reduce venetoclax with CYP3A inhibitors
† Salvage	FLAG-IDA · MEC · CLAG-M · clinical trial · second transplant

† MRD AND THE TRANSPLANT DECISION

- **Measure MRD at the end of every cycle** and again before transplant — order **HemeSTAMP MRD** (NGS) + **multiparameter flow cytometry**; add **NPM1 PCR** where applicable
- **MRD-positive CR is not CR for planning purposes** — it predicts relapse and moves an otherwise favorable-risk patient toward transplant
- **Allo-HSCT in CR1**: generally **not** for favorable risk · **yes** for intermediate and adverse risk · modified by MRD, age, comorbidity, and donor availability
- Allo cures through **graft-versus-leukemia**; **autologous transplant is not used** — the graft carries the same mutations
- Fitness for intensive therapy is a composite of age, performance status and comorbidity (HCT-CI) — not age alone

THE THREE EMERGENCIES

APL — drop everything

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Recognize
DIC + bleeding + hypergranular promyelocytes
faggot cells (bundled Auer rods)
microgranular variant can look monocytic

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Start ATRA immediately
do **not** wait for PML::RARA confirmation
early death from hemorrhage is the leading cause of APL mortality — and preventable

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Support aggressively
fibrinogen >100–150 (cryoprecipitate)
platelets >30–50k; INR near normal
coags q6–12 h; avoid LP and central lines

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Then treat
ATRA + arsenic, chemo-free, low/int risk
high risk (WBC >10k): add anthracycline
cure >90%; **do NOT leukapheresis — worsens DIC**

Leukostasis

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Hyperleukocytosis + symptoms
myeloid blasts are large and sticky — symptoms at lower counts than ALL/CLL, often >50–100k
CNS (headache, confusion, vision) + **pulmonary** (hypoxia)

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† Management — with doses
hydroxyurea 50–100 mg/kg/day PO, divided q6–12h (≈2–3 g q12h) until WBC <25k; **mucositis ≥75 mg/kg/day HU** takes days — if too slow, **cytarabine 100 mg/m²/day**
urgent induction is definitive; apheresis if severe — **never in APL**, no mortality benefit. **No RBC transfusion**
hydrate + TLS ppx. † **dex 10 mg IV q6h** if monocytic + ARF

Tumor lysis

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Cairo-Bishop
lab TLS = ≥2 of ↑K, ↑phosphate, ↑uric acid, ↓Ca
clinical TLS = lab TLS + AKI, arrhythmia or seizure

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Prevent and treat
aggressive IV hydration is the cornerstone
allopurinol 300 mg BID standard risk
rasburicase if high risk or uric acid >8 — **not in G6PD**
no urinary alkalinization

DRUG TOXICITY & MONITORING

Agent	Watch for
Cytarabine	Pyrimidine analog. High-dose: cerebellar toxicity — cerebellar exam before every HiDAC dose, hold if abnormal; conjunctivitis → prophylactic steroid eye drops ; ara-C syndrome (fever, myalgia, rash)
Anthracyclines	Cumulative, dose-dependent cardiotoxicity — track total dose, baseline ± serial LVEF. Vesicant — central access
Venetoclax	Tumor lysis on ramp-up : neutropenia; dose-reduce with azole antifungals (CYP3A)
Azacitidine vs decitabine	Both hypomethylating; azacitidine incorporates into RNA and DNA , decitabine DNA-only. Azacitidine: injection-site reactions
Gemtuzumab	Anti-CD33 calicheamicin ADC — hepatotoxicity and VOD/SOS , especially peri-transplant; fractionated dosing lowers risk
Arsenic trioxide	QT prolongation — monitor ECG, replete K and Mg
Differentiation syndrome	ATRA/ATO and also IDH and menin inhibitors . Fever, dyspnea, hypoxia, infiltrates, effusions, weight gain, AKI → dexamethasone 10 mg IV BID at first suspicion
Menin inhibitors	Revumenib, ziftomenib — differentiation syndrome + QT prolongation

SUPPORTIVE CARE

- **Transfusion**: platelets >10k (>20–50k if bleeding, procedure, or APL); **irradiated and leukoreduced** products
- † **Prophylaxis during induction**: antibacterial (levofloxacin), **antifungal (posaconazole)**, antiviral (acyclovir), PJP coverage — remember posaconazole forces a venetoclax dose reduction
- **Febrile neutropenia**: fever ≥38.3°C once (or ≥38.0°C for 1 h) with ANC <500 → **antipseudomonal beta-lactam within 1 hour**. Add vancomycin for instability, line infection, skin/soft tissue, severe mucositis or pneumonia. Antifungal if fever persists >4–7 days
- Fertility counseling before chemotherapy; central venous access; early palliative care in older or adverse-risk disease

TRIAL EVIDENCE

RATIFY	Midostaurin + 7+3 in FLT3-mutated — survival benefit
QuANTUM-First	Quizartinib + 7+3 in FLT3-ITD
ALFA-0701	Fractionated gemtuzumab + 7+3 in CD33-positive disease
E1900	Daunorubicin 90 vs 45 mg/m² — better CR and OS in age <60
VIALE-A	Azacitidine + venetoclax vs azacitidine — OS 14.7 vs 9.6 months (DiNardo, NEJM 2020)
† PARADIGM	Aza+ven vs intensive induction in fit patients (CBF/FLT3/ NPM1 <60 excluded, 72% ELN adverse): EFS 14.5 vs 6.2 mo , HR 0.57; less infection and bleeding. Phase 2; OS not significant (Fathi, NEJM 2026)
AGILE	Ivosidenib + azacitidine in IDH1-mutant unfit patients
ADMIRAL	Gilteritinib in relapsed FLT3-mutated AML — survival benefit
APL0406	ATRA + arsenic beat ATRA + chemo in low/intermediate-risk APL (Lo-Coco, NEJM 2013)
† QUAZAR AML-001	Oral azacitidine (CC-486) maintenance in CR1 — OS 24.7 vs 14.8 months
AML15	IDAC vs HiDAC consolidation — no overall survival difference
SWOG S1203	IA regimen showed no benefit and more toxicity — 7+3 remains standard

GENETICS — FINE POINTS

inv(3) / t(3;3)	MECOM (EVI1, 3q26) activation; classically normal or high platelets ; adverse and venetoclax-resistant
t(6;9)	DEK::NUP214 — associates with FLT3-ITD and basophilia ; adverse
NPM1 flow mimic	NPM1-mutated AML can be HLA-DR– / CD34– and mimic APL — confirm PML::RARA
Therapy-related	After alkylators/RT : del(5q)/del(7q) ± TP53, latency ~5–7 yr. After topoisomerase-II : 11q23 KMT2A , latency ~2–3 yr
† Germline	DDX41, RUNX1, CEBPA, GATA2 , telomere biology — send germline testing if young, family history, or multiple cytopenias; it changes related-donor selection